# STAT 5701

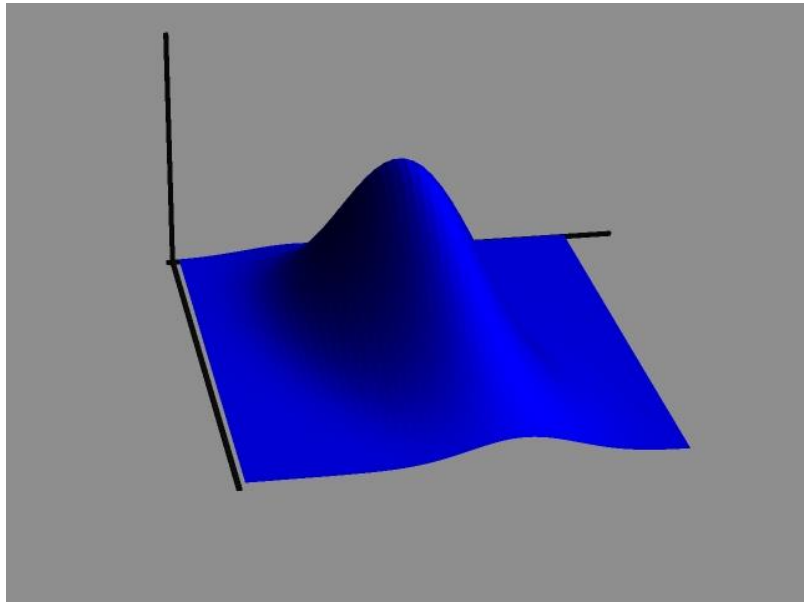
Bivariate Continuous Distributions

## Definition:

Two random variable are said to have a joint probability density function $f(x,y)$ if

1. $0 \le f(x,y)$
2. $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dxdy = 1$
3. $P\left[(X,Y) \in A\right] = \iint_{A} f(x,y)\,dxdy$

Note: $z = f(x, y)$ defines a *surface* over the $(x, y)$ plane



# Multiple Integration

Suppose we are given a positive function $f(x,y)$, defined in the domain $(x, y) \in R$.

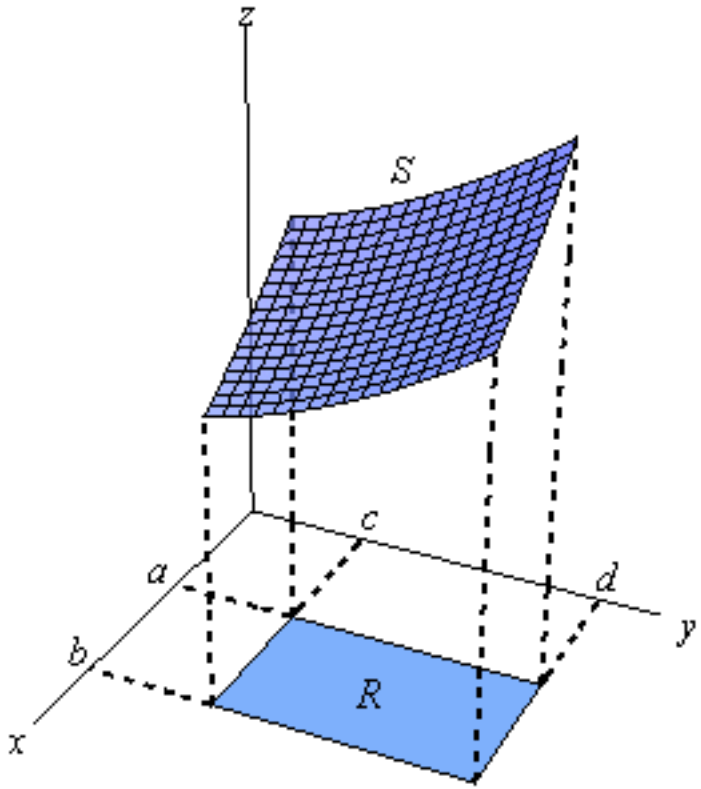
It defines a surface $S$ above the domain $R$.

The value of the double integral

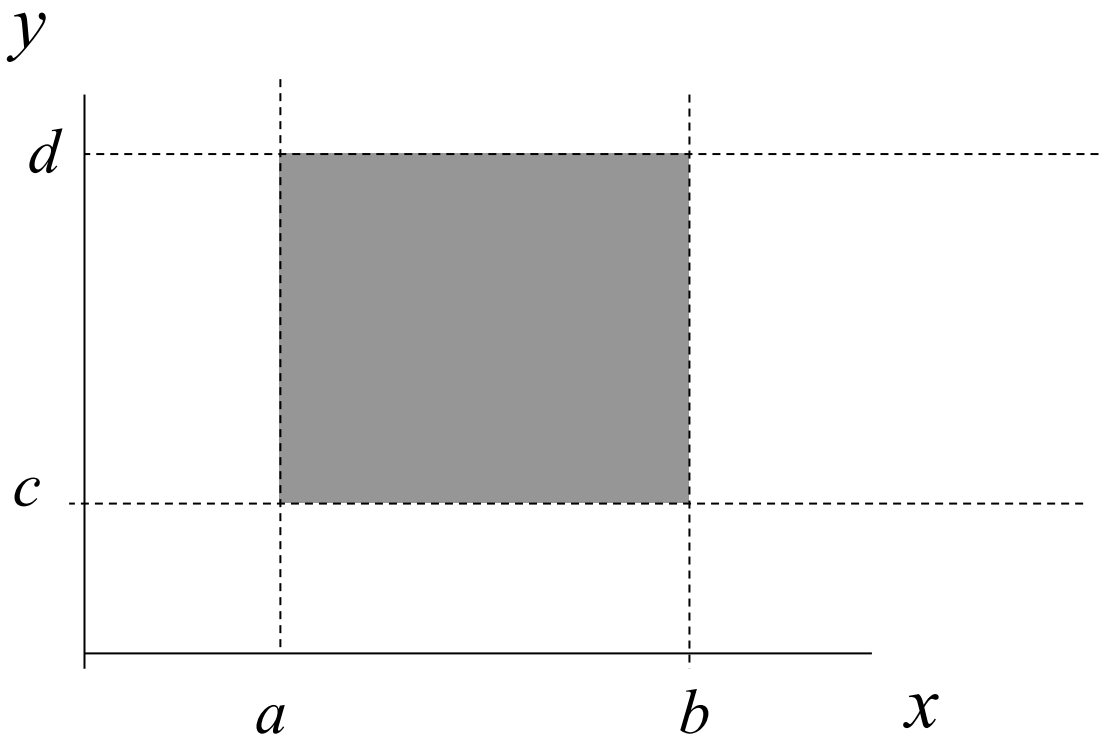
$$\iint_R f(x, y)dxdy$$

can be interpreted as the volume under the surface $S$

restricted to the domain $R$.



If the region $A = \{(x,y) | \ a \le x \le b, \ c \le y \le d\}$ is a rectangular region with sides parallel to the coordinate axes:




Then

$$\iint_A f(x,y)\,dxdy = \int_c^d \left[ \int_a^b f(x,y)\,dx \right] dy = \int_a^b \left[ \int_c^d f(x,y)\,dy \right] dx$$

To evaluate

$$\iint_A f(x,y)\,dxdy = \int_c^d \int_a^b f(x,y)\,dxdy$$

$$= \int_c^d \left[ \int_a^b f(x,y)\,dx \right] dy$$

First evaluate the *inner* integral

$$G(y) = \int_a^b f(x,y)\,dx$$

Then evaluate the *outer* integral

$$\int_c^d \int_a^b f(x,y)\,dxdy = \int_c^d G(y)\,dy$$

The same quantity can be calculated by integrating first with respect to *y,* than *x*.

$$\iint_A f(x,y)\,dxdy = \int_a^b \int_c^d f(x,y)\,dydx$$

$$= \int_a^b \left[ \int_c^d f(x,y)\,dy \right] dx$$

First evaluate the *inner* integral

$$H(x) = \int_c^d f(x,y)\,dy$$

Then evaluate the *outer* integral

$$\int_a^b \int_c^d f(x,y)\,dydx = \int_a^b H(x)\,dx$$

**Example 1:** Compute the following two integrals and check the answers are the same:

$$\int_0^1 \int_0^1 (x^2y + xy^3)dxdy = \int_0^1 \int_0^1 (x^2y + xy^3)dydx$$

*Solution:* Let's start with the lhs:

$$\int_0^1 \int_0^1 (x^2y + xy^3)dxdy = \int_0^1 \left[\int_0^1 (x^2y + xy^3)dx\right] dy$$

$$= \int_0^1 \left[\frac{x^3}{3}y + \frac{x^2}{2}y^3 \Big|_{\substack{x=1\\x=0}}\right] dy$$

$$= \int_0^1 \left(\frac{1}{3}y + \frac{1}{2}y^3\right) dy = \frac{1}{3}\frac{y^2}{2} + \frac{1}{2}\frac{y^4}{4}\Big|_{y=0}^{y=1}$$

$$= \frac{1}{6} + \frac{1}{8} = \frac{7}{24}$$

The same quantity can be computed by reversing the order of integration

$$\int_0^1 \int_0^1 \left( x^2 y + x y^3 \right) dy dx = \int_0^1 \left[ \int_0^1 \left( x^2 y + x y^3 \right) dy \right] dx$$

$$= \int_0^1 \left[ \frac{y^2}{2} x^2 + \frac{y^4}{4} x \Big|_{y=0}^{y=1} \right] dx$$

$$= \int_0^1 \left( \frac{1}{2} x^2 + \frac{1}{4} x \right) dx = \frac{1}{2} \frac{x^3}{3} + \frac{1}{4} \frac{x^2}{2} \Big|_{x=0}^{x=1}$$

$$= \frac{1}{6} + \frac{1}{8} = \frac{7}{24}$$

**Exercise:**

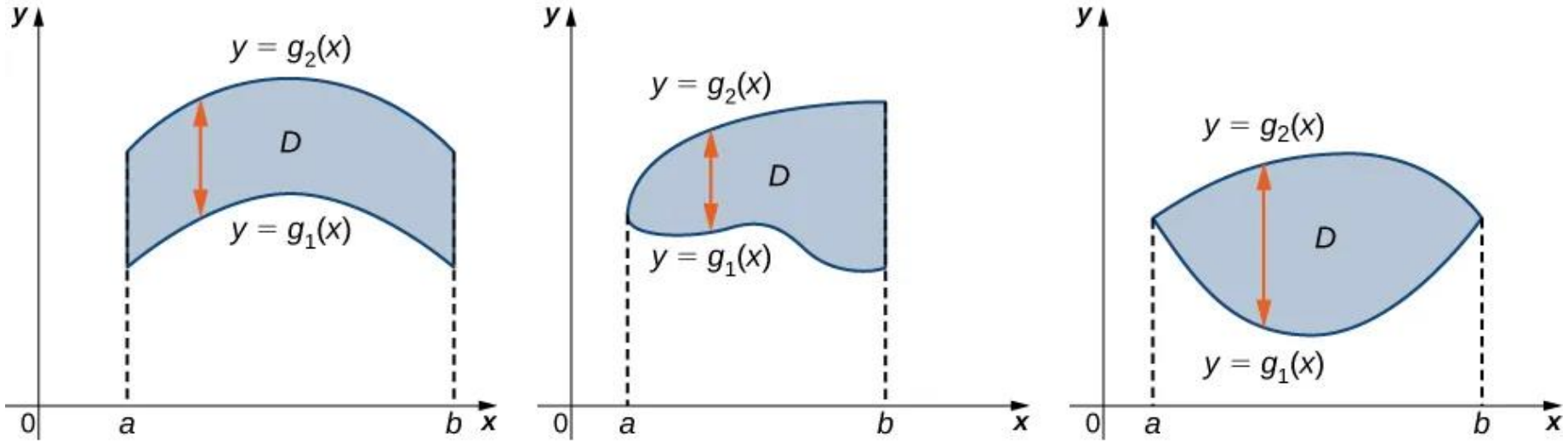
Compute

$$\int_0^2 \int_0^1 xy^2 dydx$$

# Integration over nonrectangular regions

**Type I (Vertical)**: If the region $A$ is defined as follows

$$A = \{(x,y) |\ a \le x \le b,\ g_1(x) \le y \le g_2(x)\ \}$$

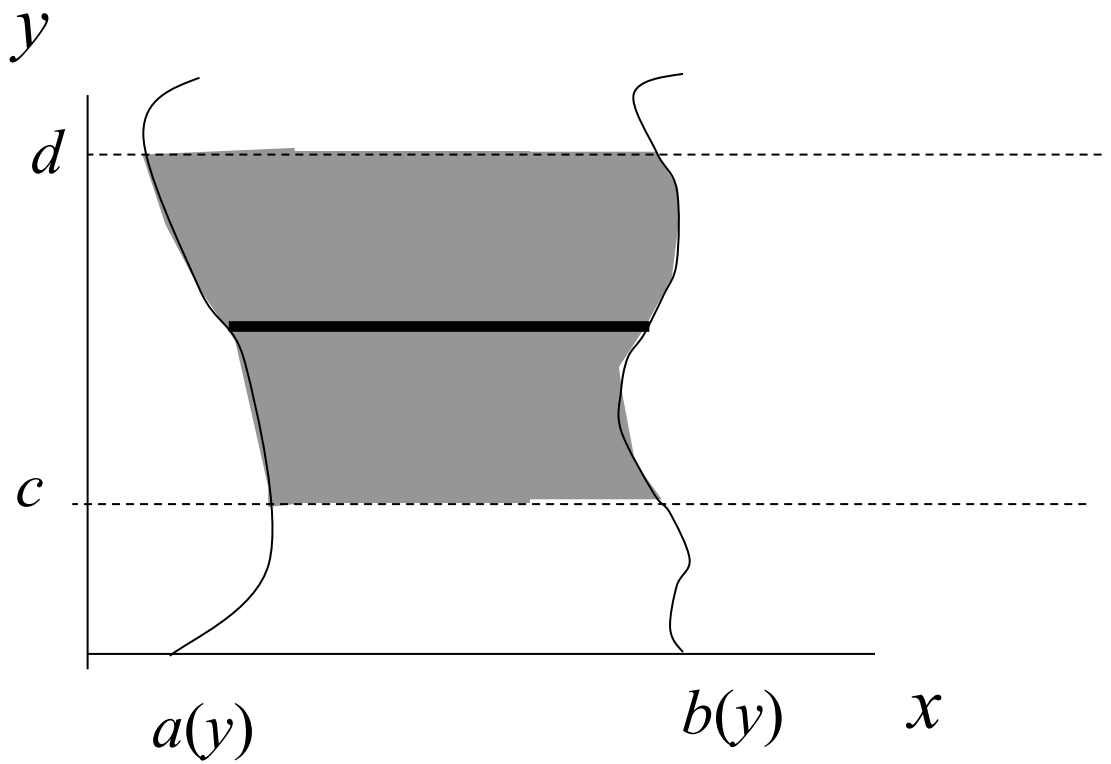



Then

$$\iint_A f(x,y)\,dxdy = \int_a^b \left[ \int_{g_1(x)}^{g_2(x)} f(x,y)\,dy \right] dx$$

Type II: Suppose the region $A$ is defined as follows

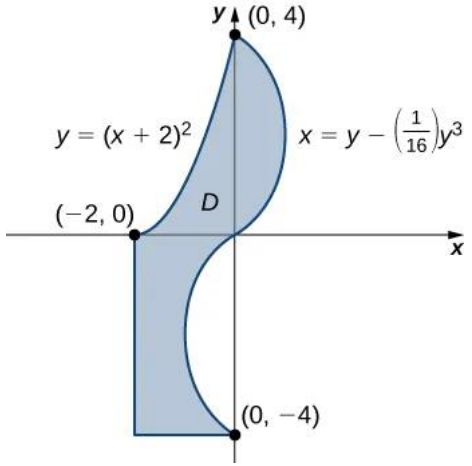
$$A = \{(x,y) \mid a(y) \le x \le b(y),\ c \le y \le d\}$$




Then

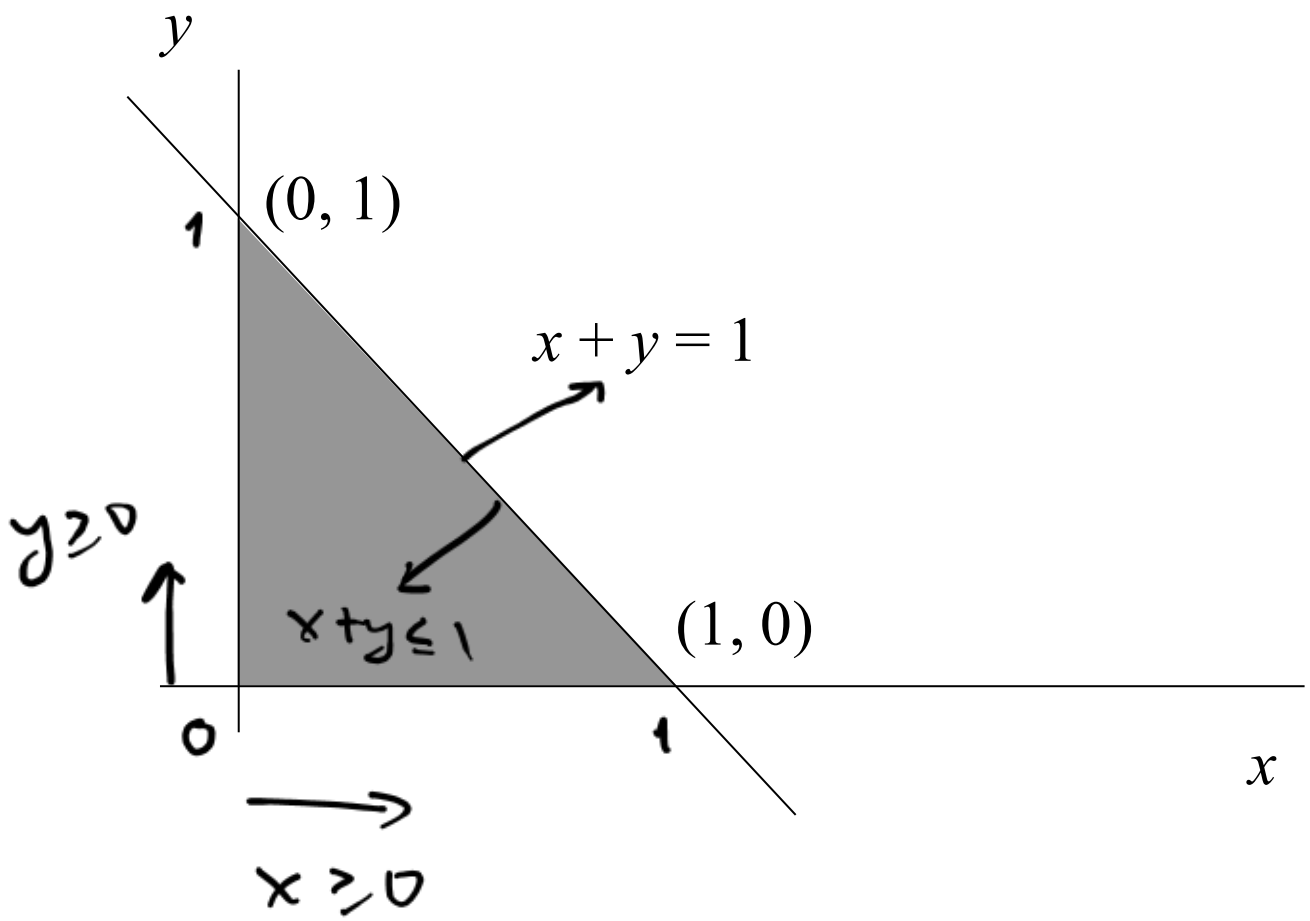
$$\iint_A f(x,y)\,dx\,dy = \int_c^d \left[ \int_{a(y)}^{b(y)} f(x,y)\,dx \right] dy$$

In general, the region $A$ can be partitioned into the two types of regions:

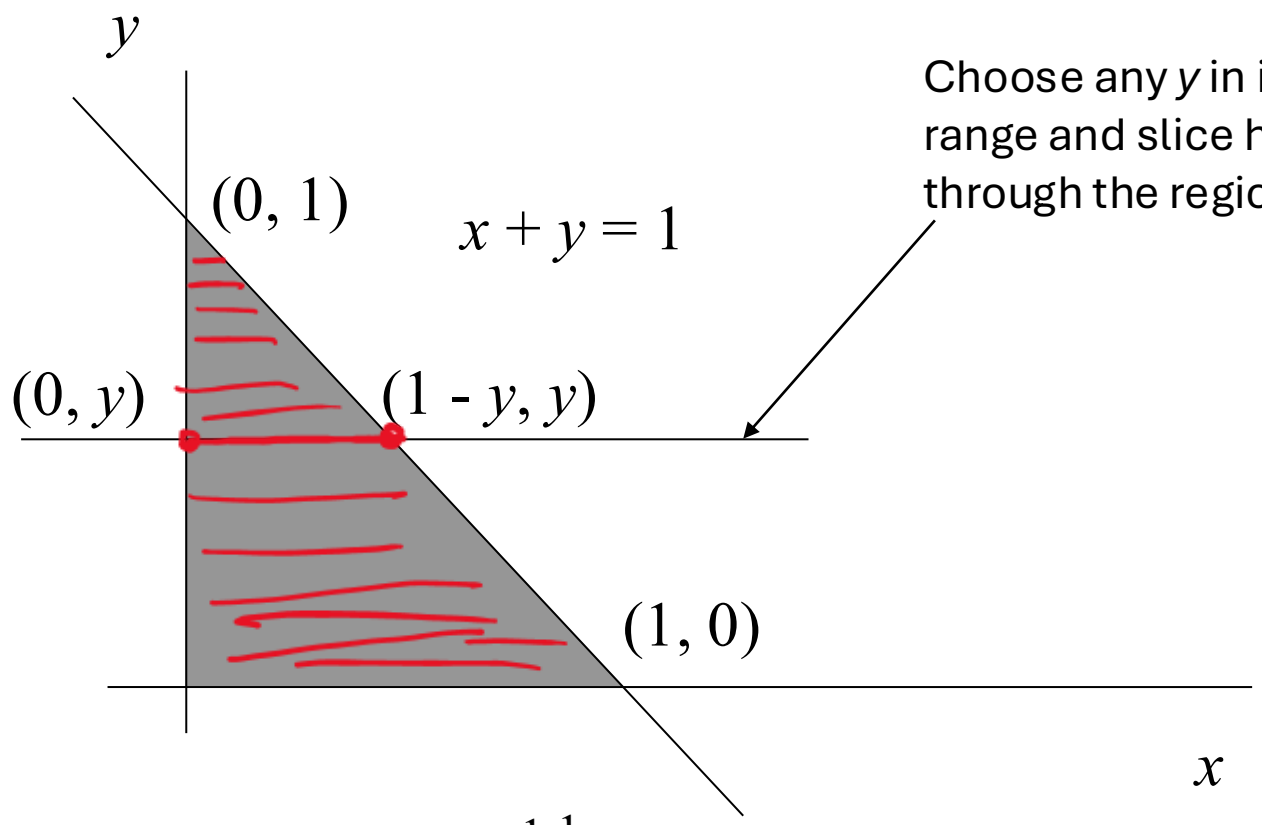


**Example 2:**

Compute the volume under $f(x,y) = x^2y + xy^3$ over the region $A = \{(x,y) | \, x + y \leq 1, \, 0 \leq x, \, 0 \leq y\}$



Integrating first with respect to $x$ for a given $y$
This means $y$ is the outer and $x$ is the inner integral.




Choose any $y$ in its allowed range and slice horizontally through the region.

$$\iint_A \left(x^2y + xy^3\right)dxdy = \int_0^1 \int_0^{1-y} \left(x^2y + xy^3\right)dxdy$$

and

$$\int_0^1 \left[ \int_0^{1-y} \left( x^2 y + x y^3 \right) dx \right] dy = \int_0^1 \left[ \frac{x^3}{3} y + \frac{x^2}{2} y^3 \Big|_{x=0}^{x=1-y} \right] dy$$

$$= \int_0^1 \left[ \frac{(1-y)^3}{3} y + \frac{(1-y)^2}{2} y^3 \right] dy$$

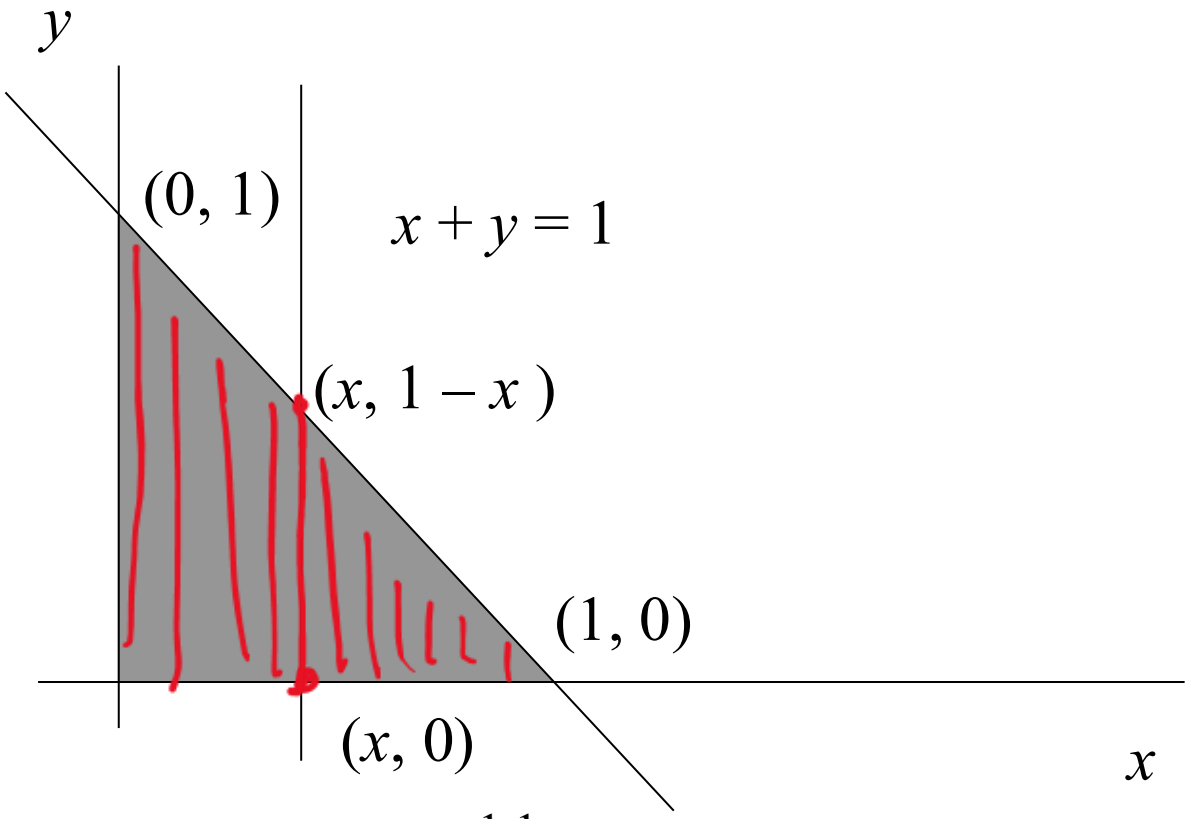
$$= \int_0^1 \left[ \frac{y - 3y^2 + 3y^3 - y^4}{3} + \frac{y^3 - 2y^4 + y^5}{2} \right] dy$$

$$= \frac{\frac{1}{2} - 1 + \frac{3}{4} - \frac{1}{5}}{3} + \frac{\frac{1}{4} - \frac{2}{5} + \frac{1}{6}}{2}$$

$$= \frac{1}{6} - \frac{1}{3} + \frac{1}{4} - \frac{1}{15} + \frac{1}{8} - \frac{1}{5} + \frac{1}{12}$$

$$= \frac{20 - 40 + 30 - 8 + 15 - 24 + 10}{120} = \frac{3}{120} = \frac{1}{40}$$

Now integrating first with respect to $y$ than $x$




$$\iint_A \left(x^2y + xy^3\right) dydx = \int_0^1 \int_0^{1-x} \left(x^2y + xy^3\right) dydx$$

$$= \int_0^1 \left[ \int_0^{1-x} \left(x^2y + xy^3\right) dy \right] dx$$

Hence $$\int_0^1 \left[ \int_0^{1-x} \left( x^2 y + x y^3 \right) dy \right] dx = \int_0^1 \left[ \left( x^2 \frac{y^2}{2} + x \frac{y^4}{4} \right) \Big|_{y=0}^{y=1-x} \right] dx$$

$$= \int_0^1 x^2 \frac{(1-x)^2}{2} + x \frac{(1-x)^4}{4} dx$$

$$= \int_0^1 \frac{x^2 - 2x^3 + x^4}{2} + \frac{x - 4x^2 + 6x^3 - 4x^4 + x^5}{4} dx$$

$$= \int_0^1 \frac{x - 2x^2 + 2x^3 - 2x^4 + x^5}{4} dx = \frac{x^2}{8} - \frac{x^3}{6} + \frac{x^4}{8} - \frac{x^5}{10} + \frac{x^6}{24} \Big]_0^1$$

$$= \frac{1}{8} - \frac{1}{6} + \frac{1}{8} - \frac{1}{10} + \frac{1}{24} = \frac{15 - 20 + 15 - 12 + 5}{120} = \frac{3}{120} = \frac{1}{40}$$

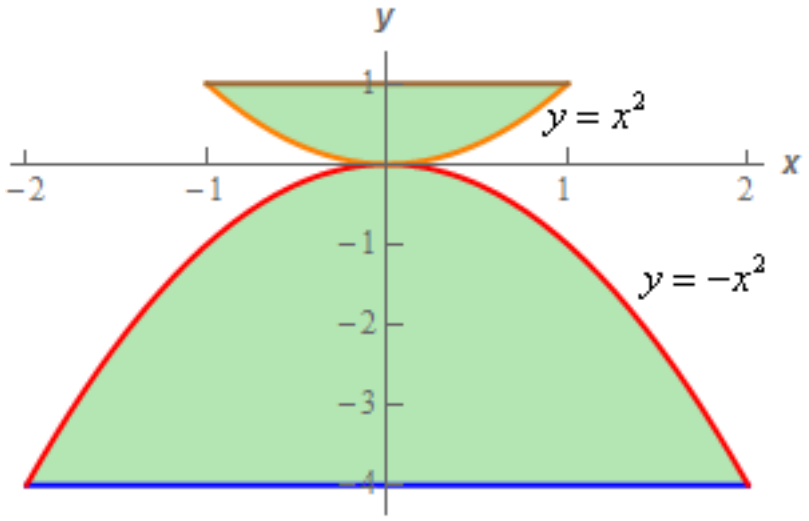
# Exercise:

Evaluate

a) the integral of $f(x, y) = 2 - 3x + xy$ over

the triangle that is spanned by (0, 0); (1, 0); (1, 3)

b) $\iint_D (3 - 6xy)dxdy$, where the region $D$ is shown below:




Answer: b) 36

# Back to Continuous Random Variables

## Example 3:

Let continuous random variables $X$ and $Y$ have an unnormalized joint density function $f(x, y) = ce^{x^2}$ over a restricted triangular domain defined by $0 \leq y \leq x \leq 1$. Compute the normalization constant.
The integration order on a non-rectangular domain ($dy\,dx$) provides the solution.

$$\int_0^1 \int_y^1 ce^{x^2}\,dx\,dy = c\int_0^1 \int_y^1 e^{x^2}\,dx\,dy = 1$$

**Attempting the original order ($dx\,dy$):** The inner integral is $\int e^{x^2}\,dx$. The function $e^{x^2}$ has no elementary antiderivative, making it impossible to evaluate directly using standard calculus techniques.

**Reversing the order of integration**

The original limits of integration describe the region $R$:

$$y \leq x \leq 1$$
$$0 \leq y \leq 1$$

This region forms a triangle with vertices at $(0,0)$, $(1,0)$, and $(1,1)$. To integrate with respect to $y$first, we must redefine the bounds for $R$ by looking at vertical cross-sections rather than horizontal ones:

$$0 \leq y \leq x$$
$$0 \leq x \leq 1$$

The new, equivalent integral becomes:

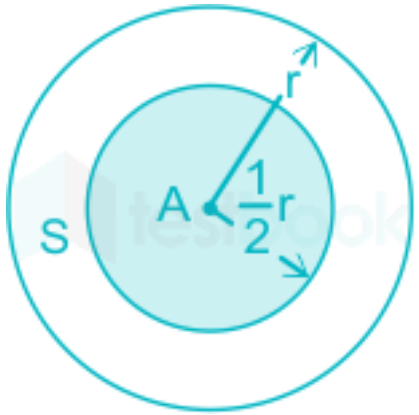
$$\int_0^1 \int_0^x e^{x^2}\,dy\,dx$$

**Example:** Let $X$ and $Y$ be random variables with joint probability density function

$$f(x,y) = \frac{1}{\pi r^2}, \qquad x^2 + y^2 \le r^2$$

That is, $(X, Y)$ is uniformly distributed in a circle with radius $R$ and center at $(0, 0)$.

Without using integrals, find the probability that $(X, Y)$ is within $\frac{r}{2}$ of the center of the circle.




$$P(A) = P\left((x,y): x^2 + y^2 \le \frac{r^2}{4}\right) = \iint_A \frac{1}{r^2}\,dxdy$$

$$= \frac{Area\ of\ A}{Area\ of\ S} = \frac{\pi\left(\frac{r}{2}\right)^2}{\pi r^2} = \frac{1}{4}$$

## Definition:

If $X$ and $Y$ are two random variables with joint probability density function $f(x, y)$, then:

- the **marginal density** of $X$ is defined as

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y)dy$$

- the **marginal density** of $Y$ is

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y)dx$$

# Definition of Conditional Density

Let $X$ and $Y$ denote two random variables with joint probability density function $f(x, y)$ and marginal densities $f_X(x)$, $f_Y(y)$ then the **conditional density** of $Y$ given $X = x$ defined as:

$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)}$$

and the **conditional density** of $X$ given $Y = y$ is:

$$f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)}$$

**Important**: Conditional densities play crucial role in data science! When we seek to model the relationship between a target variable $Y$ and one or more covariates $X$, our goal is to establish a conditional-expectation type association. This is the case of the regression models.

**Example:** Let $X$ and $Y$ be random variables with joint probability density function

$f(x, y) = (6/5)(x + y^2)$ for $0 < x < 1$, $0 < y < 1$

a) Verify this is a legitimate pdf

b) Find $P(X < \frac{1}{4}, Y < \frac{1}{4})$ $= \int_0^{1/4}\int_0^{1/4} \frac{6}{5}(x+y^2)\,dx\,dy$

$= \frac{6}{5}\int_0^{1/4} \left(\frac{x^2}{2} + y^2 x\right)\Big]_{x=0}^{1/4} dy$

$= \cdots$

c) Find $f_X(x)$ and $f_Y(y)$

d) Find $P(\frac{1}{4} < Y < \frac{3}{4})$

$= \int_{1/4}^{3/4} f_Y(y)\,dy$

$= \frac{1}{5}\int_{1/4}^{3/4} (3 + 6y^2)\,dy = \frac{37}{80}$

**Example:** Let $X$ and $Y$ be random variables with joint probability density function

$f(x, y) = (6/5)(x + y^2)$ for $0 < x < 1$, $0 < y < 1$

a) Verify this is a legitimate pdf

$$\int_0^1 \int_0^1 \left(\frac{6}{5}\right)(x + y^2)\, dx\, dy \stackrel{?}{=} 1$$

$$\frac{6}{5} \int_0^1 \left(\frac{x^2}{2} + y^2 \cdot x\right)\Big]_{x=0}^{1} dy = \frac{6}{5} \int_0^1 \left(\frac{1}{2} + y^2\right) dy$$

$$= \frac{6}{5}\left(\frac{y}{2} + \frac{y^3}{3}\right)\Big|_0^1 = \frac{6}{5}\left(\frac{1}{2} + \frac{1}{3}\right) = 1 \quad \checkmark$$

**Example:** Let $X$ and $Y$ be random variables with joint probability density function

$f(x, y) = (6/5)(x + y^2)$ for $0 < x < 1$, $0 < y < 1$

c) Find $f_X(x)$ and $f_Y(y)$




$$f(y) = \int_0^1 f(x,y)\,dx = \int_0^1 \frac{6}{5}(x+y^2)\,dx$$

$$= \frac{6}{5}\left(\frac{x^2}{2} + y^2 x\right)\Big|_{x=0}^{1} = \frac{6}{5}\left(\frac{1}{2} + y^2\right)$$

$$= \frac{3}{5} + \frac{6y^2}{5} = \frac{3+6y^2}{5}$$

$$\Rightarrow \boxed{f_Y(y) = \frac{3+6y^2}{5} \text{ for } y \in (0,1)}$$

# Example:

Let $X$ and $Y$ denote two random variables with joint probability density function $f(x, y) = cx + 1$ for $x, y > 0$ and $x + y < 1$

Find

*a)* $c$

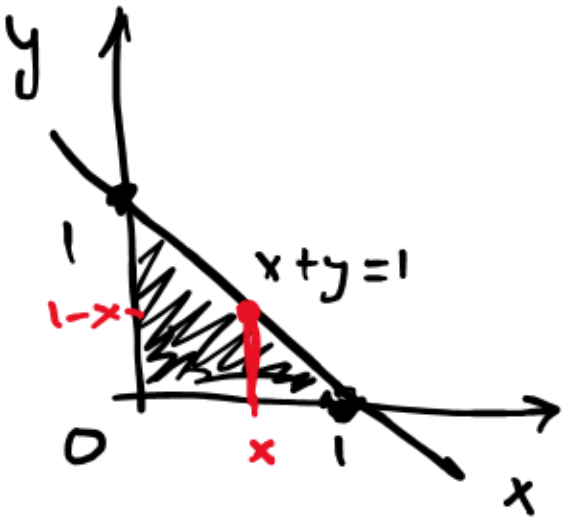
*b)* $f_X(x), f_Y(y)$

*c)* $f(x \mid y)$ and $f(y \mid x)$

**Example:** Let $X$ and $Y$ denote two random variables with joint probability density function $f(x, y) = cx + 1$ for $x, y > 0$ and $x + y < 1$

Find

*a)* $c$




$$\int_0^1 \int_0^{1-x} (cx+1)\, dy\, dx$$

$$= \int_0^1 (cx+1) \cdot y \Big|_{y=0}^{1-x} dx$$

$$= \int_0^1 (cx+1)(1-x)\, dx = \int_0^1 (cx+1-cx^2-x)\, dx$$

$$= -\frac{cx^3}{3} + (c-1)\frac{x^2}{2} + x \Big|_0^1$$

$$-\frac{c}{3} + \frac{c-1}{2} + 1 = 1$$

$c = 3$

# Example: job interview question

Suppose you choose at random two chords on a circle.

What is the probability that the chords will intersect?

# Bivariate Normal Distribution

If the random vector $(X_1, X_2)$ has a pdf

$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-\frac{1}{2}Q(x_1,x_2)}$$

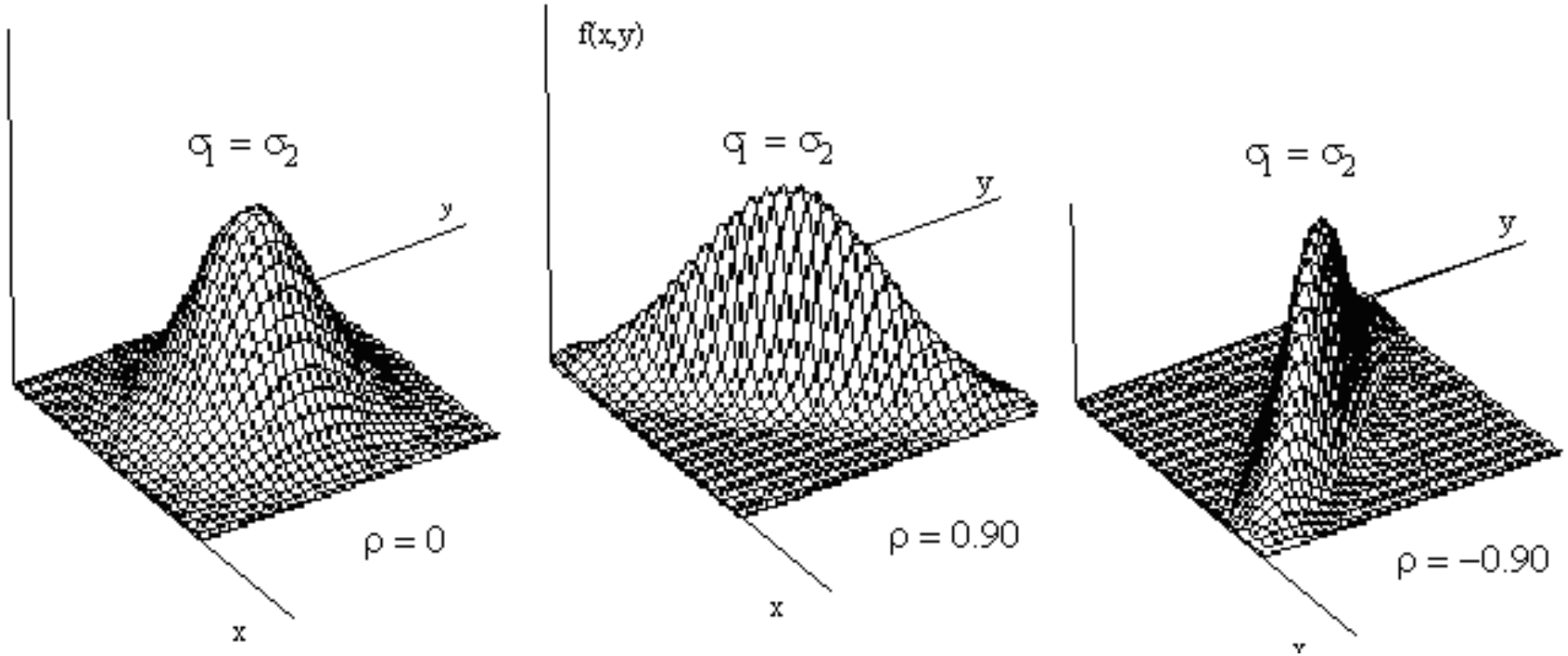
where

$$Q(x_1, x_2) = \frac{\left(\frac{x_1-\mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x_1-\mu_1}{\sigma_1}\right)\left(\frac{x_2-\mu_2}{\sigma_2}\right) + \left(\frac{x_2-\mu_2}{\sigma_2}\right)^2}{1-\rho^2}$$

Then the distribution is called the **bivariate Normal distribution.**

The parameters are $\mu_1$, $\mu_2$ , $\sigma_1$, $\sigma_2$ and $\rho$.

# Surface Plots of the bivariate Normal distribution



Note:

$$f\left(x_1, x_2\right) = \frac{1}{\left(2\pi\right)\sigma_1\sigma_2\sqrt{1-\rho^2}} \mathrm{e}^{-\frac{1}{2}Q\left(x_1, x_2\right)}$$

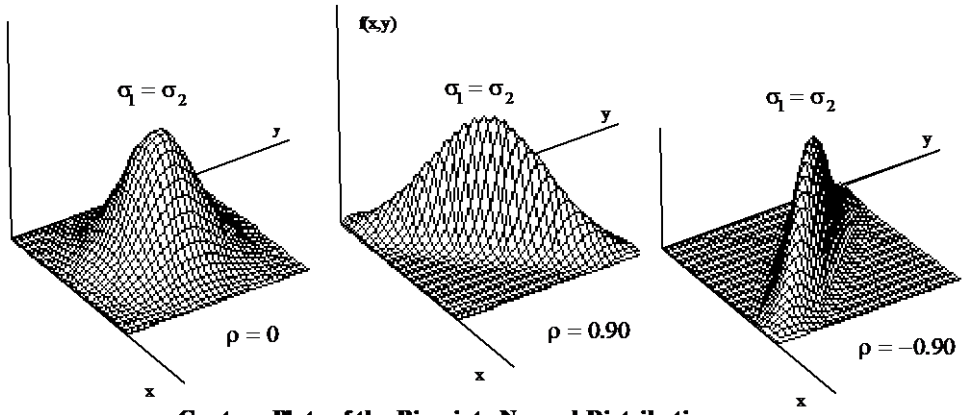
is constant when

$$Q\left(x_1, x_2\right) = \frac{\left\{\left(\frac{x_1-\mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x_1-\mu_1}{\sigma_1}\right)\left(\frac{x_2-\mu_2}{\sigma_2}\right) + \left(\frac{x_2-\mu_2}{\sigma_2}\right)^2\right\}}{1-\rho^2}$$

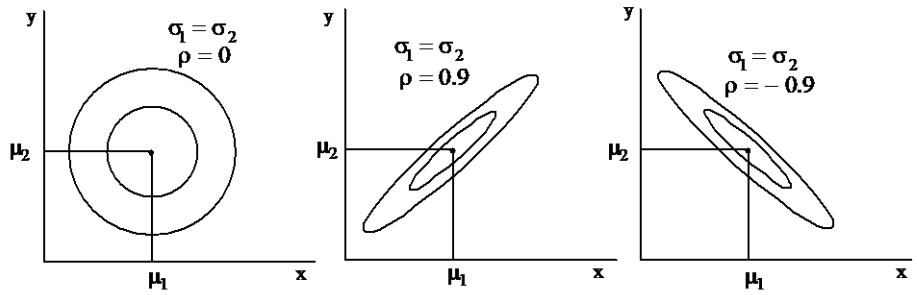
is constant.

This is true when $x_1$, $x_2$ lie on an ellipse centered at $\mu_1$, $\mu_2$ .

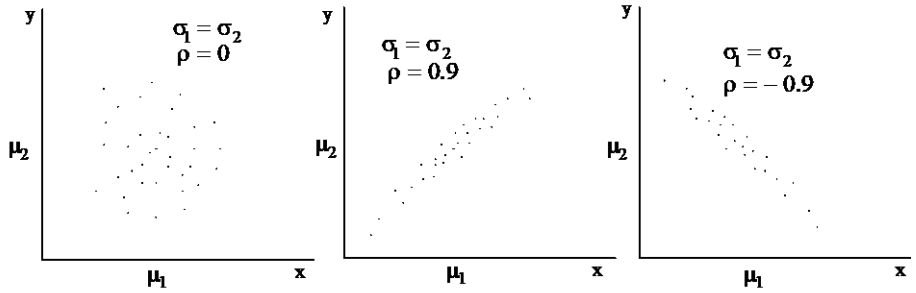
## The Bivariate Normal Distribution




## Contour Plots of the Bivariate Normal Distribution



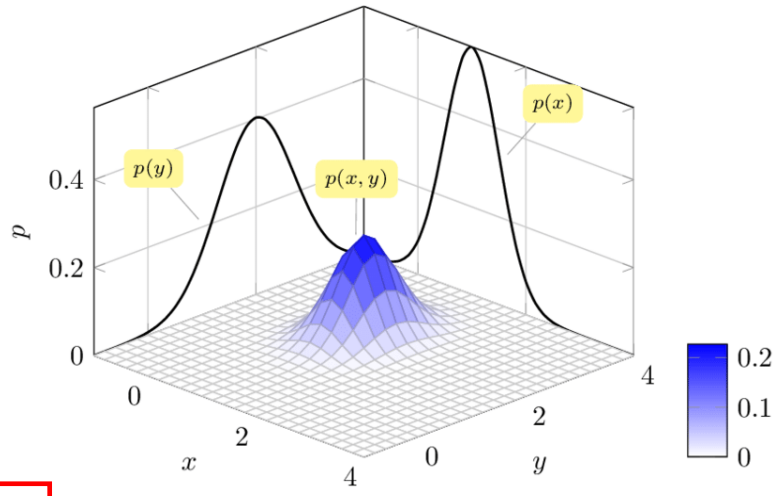

## Scatter Plots of data from the Bivariate Normal Distribution



# Marginal Distributions for the Bivariate Normal distribution

It can be shown that in the case of the bivariate normal distribution the marginal distribution of $x_i$ is Normal with mean $\mu_i$ and standard deviation $\sigma_i$, $i = 1, 2$

Thus, the marginal distribution of $x_2$ is Normal with mean $\mu_2$ and standard deviation $\sigma_2$.



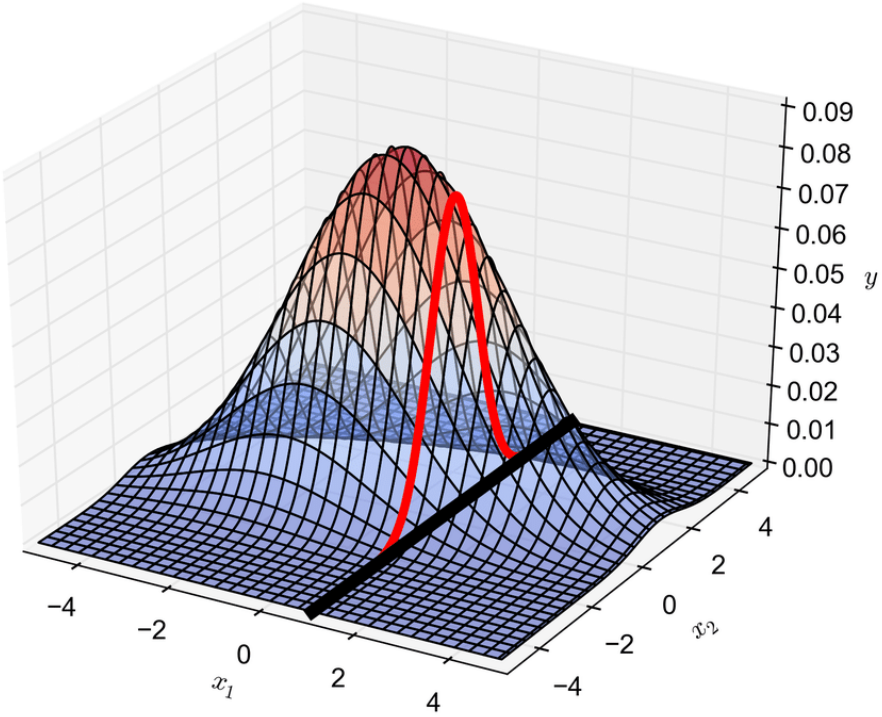

Similarly, the marginal distribution of $x_1$ is Normal with mean $\mu_1$ and standard deviation $\sigma_1$.

# Conditional Distributions of the Bivariate Normal Distribution

If $(X_1, X_2)$ follows the bivariate normal distribution, it can be shown that the conditional distributions of $X_i$ given $X_j$ , $i, j = 1, 2$ is normal with:

mean $\mu_{i|j} = \mu_i + \rho \frac{\sigma_i}{\sigma_j}\left(x_j - \mu_j\right)$

standard deviation $\sigma_{i|j} = \sigma_i \sqrt{1-\rho^2}$



# Proof

$$f_{2|1}\left(x_2 \middle| x_1\right) = \frac{f\left(x_1, x_2\right)}{f_1\left(x_1\right)}$$

$$= \frac{\dfrac{\mathrm{e}^{-\frac{1}{2}Q(x_1,x_2)}}{(2\pi)\sigma_1\sigma_2\sqrt{1-\rho^2}}}{\dfrac{1}{\sqrt{2\pi}\sigma_2} e^{-\frac{1}{2}\left(\frac{x_2-\mu_2}{\sigma_2}\right)^2}}$$

$$= \frac{\mathrm{e}^{-\frac{1}{2}Q(x_1,x_2)-\frac{1}{2}\left(\frac{x_2-\mu_2}{\sigma_2}\right)^2}}{\sqrt{2\pi}\sigma_1\sqrt{1-\rho^2}} = \frac{\mathrm{e}^{-\frac{1}{2}\left[\left(\frac{x_1-a}{b}\right)^2+c\right]-\frac{1}{2}\left(\frac{x_2-\mu_2}{\sigma_2}\right)^2}}{\sqrt{2\pi}\sigma_1\sqrt{1-\rho^2}}$$

where $b = \sigma_1 \sqrt{1-\rho^2}$

$$a = \mu_1 + \rho \frac{\sigma_1}{\sigma_2}\left(x_2 - \mu\right)$$

and $$c = \left( \frac{x_2 - \mu_2}{\sigma_2} \right)^2$$

Hence $$f_{1|2}\left(x_1 \middle| x_2\right) = \frac{1}{\sqrt{2\pi}b}\ e^{-\frac{1}{2}\left(\frac{x_1 - a}{b}\right)^2}$$

Thus the conditional distribution of $x_2$ given $x_1$ is Normal with:

mean $a = \mu_{1|2} = \mu_1 + \rho \frac{\sigma_1}{\sigma_2}\left(x_2 - \mu_2\right)$ and

standard deviation $b = \sigma_{1|2} = \sigma_1 \sqrt{1-\rho^2}$

# Multivariate distributions

$k \geq 2$

# The multivariate normal distribution

The pdf of the $k$-variate normal is:

$$f(x_1, \ldots, x_k) = f(\boldsymbol{x}) = \frac{1}{(2\pi)^{\frac{k}{2}}\sqrt{\det(\boldsymbol{\Sigma})}} e^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})}$$

where

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \mathrm{M} \\ x_k \end{bmatrix} \qquad \boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mathrm{M} \\ \mu_k \end{bmatrix} \qquad \Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \mathrm{L} & \sigma_{1k} \\ \sigma_{12} & \sigma_{22} & \mathrm{L} & \sigma_{2k} \\ \mathrm{M} & \mathrm{M} & \mathrm{O} & \mathrm{M} \\ \sigma_{1k} & \sigma_{2k} & \mathrm{L} & \sigma_{kk} \end{bmatrix}$$

# Definition: Multivariate distribution

Let $X_1, X_2, \ldots, X_n$ denote $n$ continuous random variables, then

$$f(x_1, x_2, \ldots, x_n)$$

is joint density function of $X_1, X_2, \ldots, X_n$ if

1. $f(x_1, \ldots, x_n) \geq 0$

2. $$\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(x_1, \ldots, x_n)\, dx_1, \ldots, dx_n = 1$$

3. $$P\left[(X_1, \ldots, X_n) \in A\right] = \int_A \cdots \int f(x_1, \ldots, x_n)\, dx_1, \ldots, dx_n$$

Note: A similar definition holds for discrete multivariate case.

# Definition – Mutual Independence

Let $X_1, X_2, \ldots, X_k$ denote $k$ continuous random variables with joint probability density function

$$f(x_1, x_2, \ldots, x_k)$$

then the variables $X_1, X_2, \ldots, X_k$ are called **mutually independent** if

$$f(x_1, \mathrm{K}, x_k) = f_1(x_1) f_2(x_2) \ldots f_k(x_k)$$

A similar definition for discrete random variables.

**Example:** *The Multinomial distribution*

Suppose that we observe independently $n$ times an experiment that has $k$ possible outcomes $\{O_1, O_2, \ldots, O_k\}$.

Let $p_1, p_2, \ldots, p_k$ denote the probabilities of $O_1, O_2, \ldots, O_k$ respectively.

Let $X_i$ denote the number of times that outcome $O_i$ occurs in the $n$ repetitions of the experiment.

Then the joint probability function of the random variables $X_1, X_2, \ldots, X_k$ is

$$p(x_1, \mathrm{K}\ , x_n) = \frac{n!}{x_1!\, x_2!\, \mathrm{K}\ x_k!}\, p_1^{x_1} p_2^{x_2}\, \mathrm{K}\ \ p_k^{x_k}$$

$$\frac{n!}{x_1!x_2!\mathrm{K}\ x_k!}=\binom{n}{x_1\quad x_2\quad \mathrm{L}\quad x_k}$$

is the number of ways of choosing the positions for the $x_1$ outcomes $O_1$, $x_2$ outcomes $O_2$, …, $x_k$ outcomes $O_k$

$$\binom{n}{x_1}\binom{n-x_1}{x_2}\binom{n-x_1-x_2}{x_3}\mathrm{K}\binom{x_k}{x_k}$$

$$=\left(\frac{n!}{x_1!(n-x_1)!}\right)\left(\frac{(n-x_1)!}{x_2!(n-x_1-x_2)!}\right)\left(\frac{(n-x_1-x_2)!}{x_3!(n-x_1-x_2-x_3)!}\right)\mathrm{K}$$

$$=\frac{n!}{x_1!x_2!\mathrm{K}\ x_k!}$$

## Summary

$$p(x_1, \ldots, x_k) = \frac{n!}{x_1! \ldots . x_k!} p_1^{x_1} \ldots p_k^{x_k}$$

$$= \binom{n}{x_1 \quad x_2 \quad \mathrm{K} \quad x_k} p_1^{x_1} p_2^{x_2} \mathrm{K}\ p_k^{x_k}$$

is called the **Multinomial** distribution

# Example:

Suppose that a treatment for back pain has three possible outcomes:

$O_1$ - Complete cure (no pain) – (30% chance)

$O_2$ - Reduced pain – (50% chance)

$O_3$ - No change – (20% chance)

Hence $p_1 = 0.30$, $p_2 = 0.50$, $p_3 = 0.20$.

Suppose the treatment is applied to $n = 4$ patients suffering back pain and let $X$ = the number that result in a complete cure, $Y$ = the number that result in just reduced pain, and $Z$ = the number that result in no change.

Find the distribution of $X$, $Y$ and $Z$. Compute $P[X + Y \geq Z]$

$$p(x,y,z) = \frac{4!}{x!y!z!}(0.30)^x(0.50)^y(0.20)^z \quad x+y+z=4$$

Table: $p(x,y,z)$

| $x$ | $y$ | $z$ | | | | |
|---|---|---|---|---|---|---|
| | | 0 | 1 | 2 | 3 | 4 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0.0016 |
| 0 | 1 | 0 | 0 | 0 | 0.0160 | 0 |
| 0 | 2 | 0 | 0 | 0.0600 | 0 | 0 |
| 0 | 3 | 0 | 0.1000 | 0 | 0 | 0 |
| 0 | 4 | 0.0625 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 | 0.0096 | 0 |
| 1 | 1 | 0 | 0 | 0.0720 | 0 | 0 |
| 1 | 2 | 0 | 0.1800 | 0 | 0 | 0 |
| 1 | 3 | 0.1500 | 0 | 0 | 0 | 0 |
| 1 | 4 | 0 | 0 | 0 | 0 | 0 |
| 2 | 0 | 0 | 0 | 0.0216 | 0 | 0 |
| 2 | 1 | 0 | 0.1080 | 0 | 0 | 0 |
| 2 | 2 | 0.1350 | 0 | 0 | 0 | 0 |
| 2 | 3 | 0 | 0 | 0 | 0 | 0 |
| 2 | 4 | 0 | 0 | 0 | 0 | 0 |
| 3 | 0 | 0 | 0.0216 | 0 | 0 | 0 |
| 3 | 1 | 0.0540 | 0 | 0 | 0 | 0 |
| 3 | 2 | 0 | 0 | 0 | 0 | 0 |
| 3 | 3 | 0 | 0 | 0 | 0 | 0 |
| 3 | 4 | 0 | 0 | 0 | 0 | 0 |
| 4 | 0 | 0.0081 | 0 | 0 | 0 | 0 |
| 4 | 1 | 0 | 0 | 0 | 0 | 0 |
| 4 | 2 | 0 | 0 | 0 | 0 | 0 |
| 4 | 3 | 0 | 0 | 0 | 0 | 0 |
| 4 | 4 | 0 | 0 | 0 | 0 | 0 |

$$P[X + Y \geq Z]$$

$$= 0.9728$$

| $x$ | $y$ | $z$ = 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0.0016 |
| 0 | 1 | 0 | 0 | 0 | 0.0160 | 0 |
| 0 | 2 | 0 | 0 | 0.0600 | 0 | 0 |
| 0 | 3 | 0 | 0.1000 | 0 | 0 | 0 |
| 0 | 4 | 0.0625 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 | 0.0096 | 0 |
| 1 | 1 | 0 | 0 | 0.0720 | 0 | 0 |
| 1 | 2 | 0 | 0.1800 | 0 | 0 | 0 |
| 1 | 3 | 0.1500 | 0 | 0 | 0 | 0 |
| 1 | 4 | 0 | 0 | 0 | 0 | 0 |
| 2 | 0 | 0 | 0 | 0.0216 | 0 | 0 |
| 2 | 1 | 0 | 0.1080 | 0 | 0 | 0 |
| 2 | 2 | 0.1350 | 0 | 0 | 0 | 0 |
| 2 | 3 | 0 | 0 | 0 | 0 | 0 |
| 2 | 4 | 0 | 0 | 0 | 0 | 0 |
| 3 | 0 | 0 | 0.0216 | 0 | 0 | 0 |
| 3 | 1 | 0.0540 | 0 | 0 | 0 | 0 |
| 3 | 2 | 0 | 0 | 0 | 0 | 0 |
| 3 | 3 | 0 | 0 | 0 | 0 | 0 |
| 3 | 4 | 0 | 0 | 0 | 0 | 0 |
| 4 | 0 | 0.0081 | 0 | 0 | 0 | 0 |
| 4 | 1 | 0 | 0 | 0 | 0 | 0 |
| 4 | 2 | 0 | 0 | 0 | 0 | 0 |
| 4 | 3 | 0 | 0 | 0 | 0 | 0 |
| 4 | 4 | 0 | 0 | 0 | 0 | 0 |

# Definition

Let $X_1, X_2, \ldots, X_q, X_{q+1} \ldots, X_n$ denote $n$ continuous random variables with joint probability density function

$$f(x_1, x_2, \ldots, x_q, x_{q+1} \ldots, x_n)$$

then the marginal joint probability function of $X_1, X_2, \ldots, X_q$ is

$$f_{12\ldots q}(x_1, \ldots, x_q) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(x_1, \ldots, x_n) dx_{q+1} \ldots dx_n$$

Note: The discrete case is defined similarly

## Definition: Conditional Distributions

Let $X_1, X_2, \ldots, X_q, X_{q+1} \ldots, X_k$ denote $k$ continuous random variables with joint probability density function

$$f(x_1, x_2, \ldots, x_q, x_{q+1} \ldots, x_k)$$

then the **conditional** joint probability function of $X_1, X_2, \ldots, X_q$ given $X_{q+1} = x_{q+1}, \ldots, X_k = x_k$ is

$$f_{1\mathrm{K}\ q|q+1\mathrm{K}\ k}\left(x_1,\ldots,x_q \,\middle|\, x_{q+1},\ldots,x_k\right) = \frac{f\left(x_1,\mathrm{K}\ ,x_k\right)}{f_{q+1\mathrm{K}\ k}\left(x_{q+1},\ldots,x_k\right)}$$